

SECTION B

7 (short)

(a) Fig. 7 shows four tensile nominal stress-strain curves up to failure for materials A, B, C and D. Each of these curves represents either a metal, a polymer, or a ceramic. In each case, identify which it is, giving reasons for your choice. [4]

(b) For the case of the metal, sketch the nominal stress-strain curve in tension, and superimpose a sketch of the corresponding true stress-strain curve in both tension and compression. [2]

(c) For the case of the ceramic, sketch the expected form of the nominal stress-strain response in both tension and compression. By sketching the failure mechanism in each case, explain the relative strength observed in tension and compression. [4]

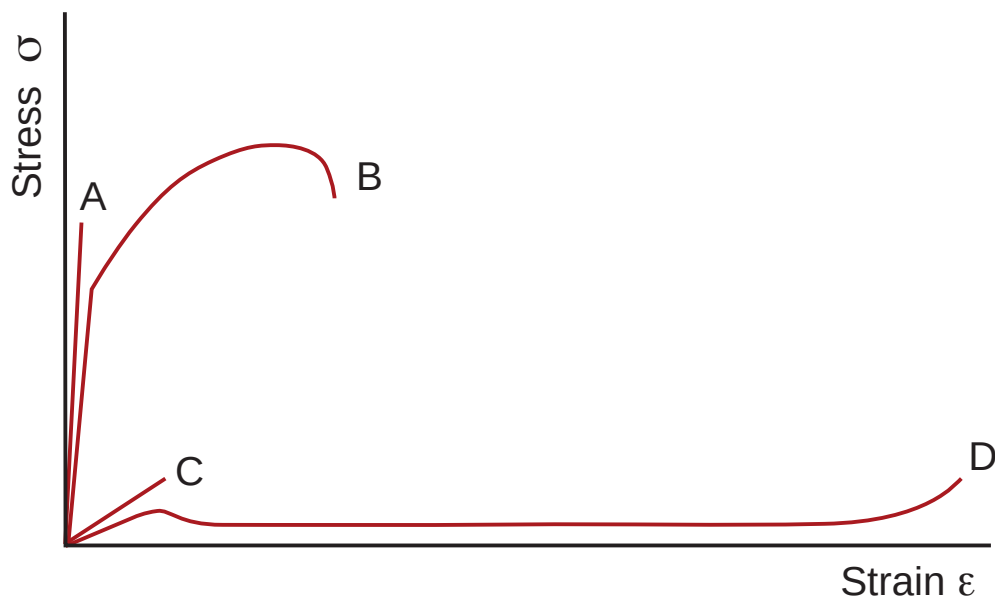


Fig. 7